

However, as technology has moved further, chip designs have evolved and in another such research initiative, a team led by Anantha Chandrakasan, the director of the research lab at MIT, has managed to develop a new chip design that they claim to be up to 10 times more energy-efficient than present technology. Ten times more energy efficient implies that if your device chips consume 100 units of electricity in a particular time period, being replaced with these new microchips they will be ten times more effective and consume a measly 10 units of power without compromising output.

The efficiency of these chips was already displayed when they were tested on a widely used microcontroller device and the results were quite promising.

Now the question that arises is that how do these chips manage to deliver the same performance at much lower power feeds? The answer to this is the ultra low level of voltage that is sufficient enough to power these new designs. These new chips eat up a mere 0.3 volt of power, which is around one-third the value of what the contemporarily implemented chips consume. The new design implements an efficient DC to DC converter that allows voltage reduction without the need for separate components, and makes it possible for it to work on such a low voltage. Chandrakasan takes it a step further by saying that the team is working to make this voltage requirement further down, so much so that the microchips would work even on the available ambient energy; viz. using one's own body heat or the kinetic energy derived from bodily movements.

According to the researchers, the new chip would hit the market and find its implementation in gadgets in a span of five years while the other issues related to its mass production and cost efficiency are being worked on. According to Chandrakasan, it would be crucial in military applications and tiny sensors where a frequent recharge value results in loss of vital time. The longevity of such devices and other mass communication and networking devices would mean an enhanced user experience as well, without worrying about frequent discharge and running around for a charging socket.

Cash On The Trash!

Moving away from computing and portables, there are significant breakthroughs being achieved in other areas of energy efficient implementations as well. The noise about exhaustible sources of energy resources and their impending death is

reverberating loud now. In the meanwhile, without making much noise, there are significant initiatives and research being carried out by individuals who are doing their small bit to protect our planet from peril.

Domestic waste is just inevitable. It is this vast resource which is being tapped by a Nigerian civil engineer Dr Joseph Adelegan—who is generating energy from domestic waste. His unparalleled credits have been awarded with adulation and due recognition for his “Cows to Kilowatts” project that generates cooking gas from waste products in abattoirs. The project today provides cooking gas supply to more than 6000 homes in Nigeria.

Imagine the vast amount of LPG (chemically butane) that would be saved if every household worldwide manages to latch on to this novel initiative. It is estimated that Nigeria accounts for over 20 per cent of the world's cassava output and it is in this waste that Dr Adelegan saw the resourceful supply of power. His project “Power to the Poor: Off-Grid Lighting from Cassava Waste in Nigeria,” is based on the principle of generating power from cassava.

The system relies on the anaerobic fixed film biogas technology that is different from existing approaches. The septic tank and soak away system and existing biogas technology use the conventional anaerobic treatment process which has several drawbacks such as extremely low efficiency and long retention time. However, the hybrid Universal Anaerobic Sludge Blanket-Anaerobic Fixed Film (UASB-AFF) bioreactor replaces the septic tank and soak away system. It also replaces conventional biogas technology by advanced high rate anaerobic reactor produced by systematic engineering design.

Using this method, Adelegan says he hopes to generate 200 kw daily, which will provide basic electricity for more than 2000 households initially.

Besides computing, technology is seeing a green revolution in other walks of life as well. The list is growing... hybrid vehicles, hydrogen fuels, solar panels and so on. The researchers at MIT are even contemplating artificial photosynthesis. These will definitely leave a long lasting impact on the technology of the future. Be it hydrogen cars or bio-diesels, unwired communication, ubiquitous computing or other aspects, it won't be too long before further groundbreaking research makes the current technology obsolete. ☒

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